VestibuLab

VIRTUAL BPPV LABORATORY

User manual

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audio Medical MX
ALTA ESPECIALIDAD AUDIOLOGÍA Y OTONEUROLOGÍA

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Important notice

VestibuLab is a learning tool developed with the support of artificial intelligence models and it can make mistakes. It is not a medical device or a clinical diagnostic tool: it must not be used to make decisions about real patients and it does not replace examination or clinical judgment.

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1 What is VestibuLab and what is it for?

VestibuLab is a virtual laboratory for benign paroxysmal positional vertigo (BPPV). It shows a patient's inner ear in three dimensions, the otoconia inside the semicircular canals and the nystagmus they produce during diagnostic tests and repositioning maneuvers. Everything is computed in real time with a physical model based on canal anatomy and the BPPV literature.

It is designed for students, residents and professionals in audiology, otoneurology, otolaryngology, neurology, vestibular physical therapy and related fields, and for teachers who want to explain BPPV in class.

What you can do

- **Free simulator:** choose one or more affected canals and their variant, run any of more than 50 tests and maneuvers, move the patient freely and watch the otoconia, the cupula, the eye and the videonystagmography (VNG) recording.
- **Training:** solve random clinical cases with the canal hidden. You run the test, observe the nystagmus and answer questions on diagnosis, mechanism, treatment and vestibular physiology.
- **Clinical video:** load a video-Frenzel recording of your patient, mark the positions of the test, track the pupil, get a probable diagnosis by comparison with the model, and watch the model patient repeat the test in sync with your video.

Editions

VestibuLab runs in the browser (it can also be installed as a web app) and as an Android app. In the launch edition all features are available free of charge. If paid features are added in the future, the purchase will be handled entirely by Google Play.

2 How it was created

VestibuLab was born from the clinical and teaching practice of **Dr. Antonio Aldana Vargas**, a physician with advanced specialty training in audiology and otoneurology and co-founder of AudioMedical MX. When teaching BPPV, the most common difficulty is picturing in three dimensions where the otoconia are, where they move in each position and why the nystagmus beats in the direction it does. The idea was to build a tool where this can be *seen* rather than memorized.

The project was developed with the support of artificial intelligence models for programming, under the clinical direction of the author, who defined each test, maneuver, variant and case, and checked the results against the literature and his experience. The application was built in stages:

1. **Anatomical model:** each canal was placed in its plane as measured by CT (Della Santina et al., 2005), with its ampullae, cupulae, common crus, utricle and saccule.
2. **Physical model:** otoconia settle according to Stokes' law in the endolymph; the signal of each canal follows Ewald's laws and the nystagmus results from the sum of the stimulated canals.
3. **Tests and maneuvers:** each was programmed step by step with the positions and times described by its authors.
4. **Clinical verification:** the training cases were run and checked twice, and the nystagmus of each variant was reviewed against the Bárány Society classification.
5. **Clinical video module:** pupil and iris-torsion tracking in video-Frenzel recordings, and its comparison with the model, were added.

The result is an educational tool: a schematic model, useful to understand BPPV, that does not replace examining the patient.

3 Getting started

How to open VestibuLab

- **In the browser:** go to *audiomedicalmx.github.io/Vestibulab* in Chrome, Edge or Safari. You can install it as an app with the browser's «Install» or «Add to home screen» option; it then opens full screen and works offline.
- **On Android:** install the VestibuLab app. Updates arrive through the store.

Registration

The first time, a short registration is requested (name, country, email, profession, institution and main use). It is used to keep a record of users and to improve the tool; details are in the «Privacy notice» chapter.

Getting-started guide

After registering, an 8-step guide points out the main parts of the screen. You can see it again with the **Guide** button or in **Settings → Getting-started guide**.

Screen layout

On a computer there are three columns: **Configuration** on the left, the **3D view** in the center and the **eye and VNG recording** on the right.

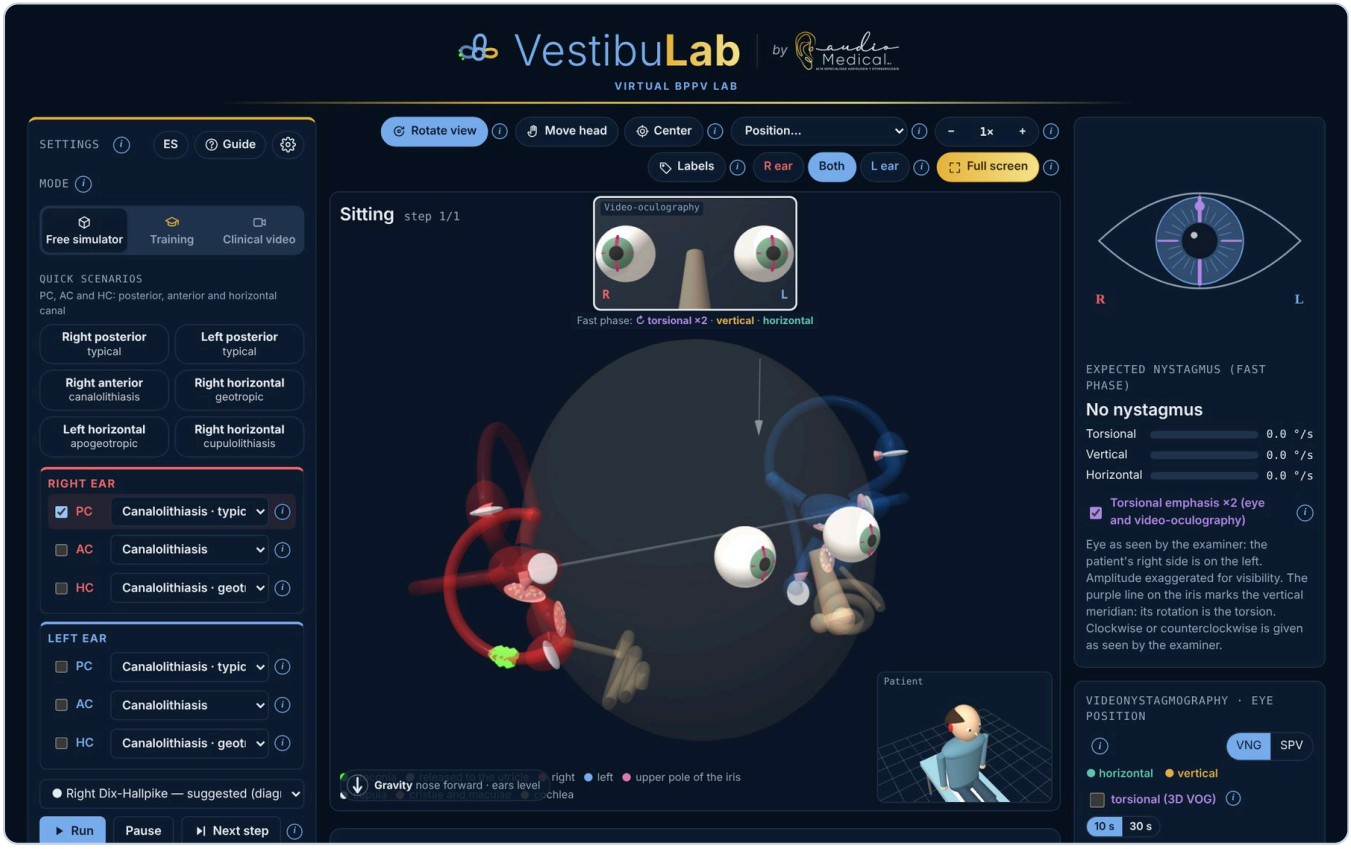


Figure 1. Main screen on a computer.

On a phone the three parts are split into tabs at the bottom: **Set up**, **Simulation** and **Eye**.



Figure 2. Set up, Simulation and Eye tabs on a phone.

Header buttons

Button	Function
EN / ES	Switches the language between English and Spanish.
Guide	Opens the quick guide.
⚙ Settings	Opens the settings («Settings» chapter).
i	The circles with «i» next to each control open a short explanation.

Work modes

At the top of the configuration panel you choose the module: **Free simulator**, **Training** or **Clinical video**.



Figure 3. Module selector.

4 Free simulator module

This is the full laboratory: you decide which canal is affected, which test or maneuver is performed and how the patient moves.

Quick scenarios

Six buttons load the most common cases in one tap: typical right and left posterior, right anterior, right horizontal geotropic, left horizontal apogeotropic and right horizontal cupulolithiasis. Each scenario selects the canal and variant and loads the suggested diagnostic test.

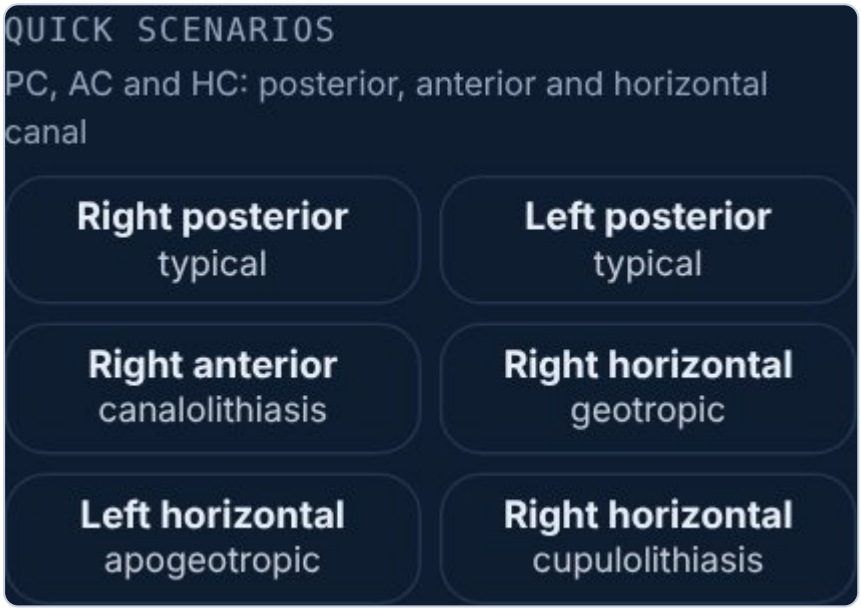


Figure 4. Quick scenarios. PC, AC and HC: posterior, anterior and horizontal canal.

Affected canal and variant

Under **Right ear** and **Left ear**, tick the canal (PC, AC or HC) and choose the variant. You can combine several canals. The «i» of each canal explains how the chosen variant behaves.

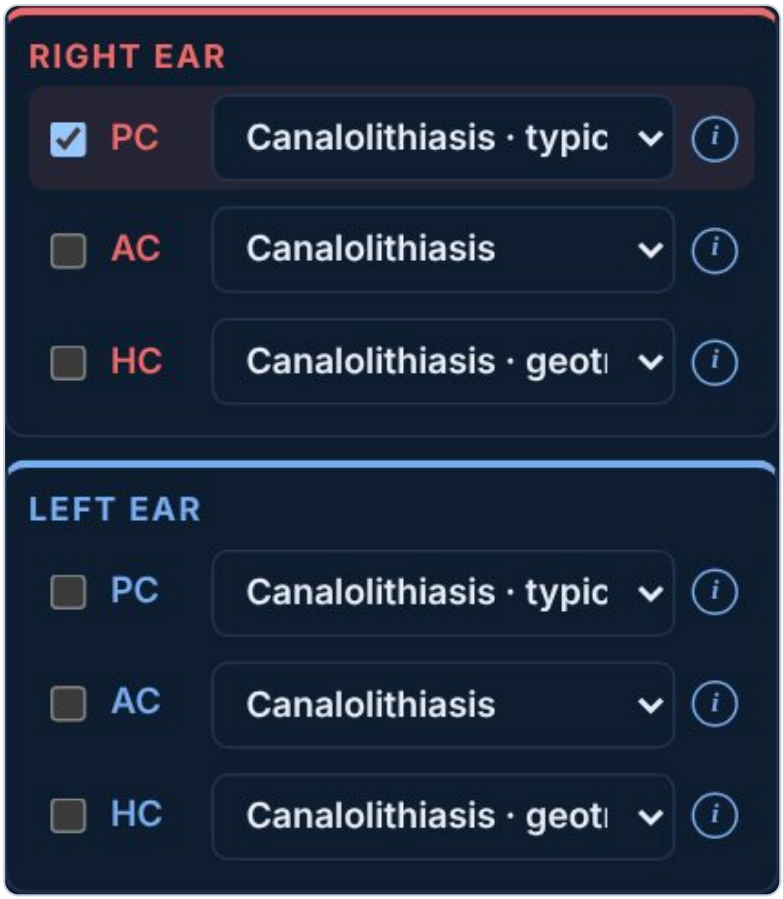


Figure 5. Choosing canals and variants.

Canal	Variants
Posterior (PC)	Typical canalolithiasis (long arm) · apogeotropic canalolithiasis (non-ampullary arm) · attached cupulolithiasis · detachable cupulolithiasis.
Anterior (AC)	Canalolithiasis · attached cupulolithiasis · detachable cupulolithiasis.
Horizontal (HC)	Geotropic canalolithiasis (posterior arm) · apogeotropic canalolithiasis (anterior arm) · attached cupulolithiasis · detachable cupulolithiasis · light cupula (differential diagnosis without otoconia).

Suggestions

For the chosen variant, the **Suggestions** box proposes the diagnostic tests that best show it and the numbered treatments (★1, ★2…) in an approximate order of efficacy, with a clinical note. Tap any of them to load it.

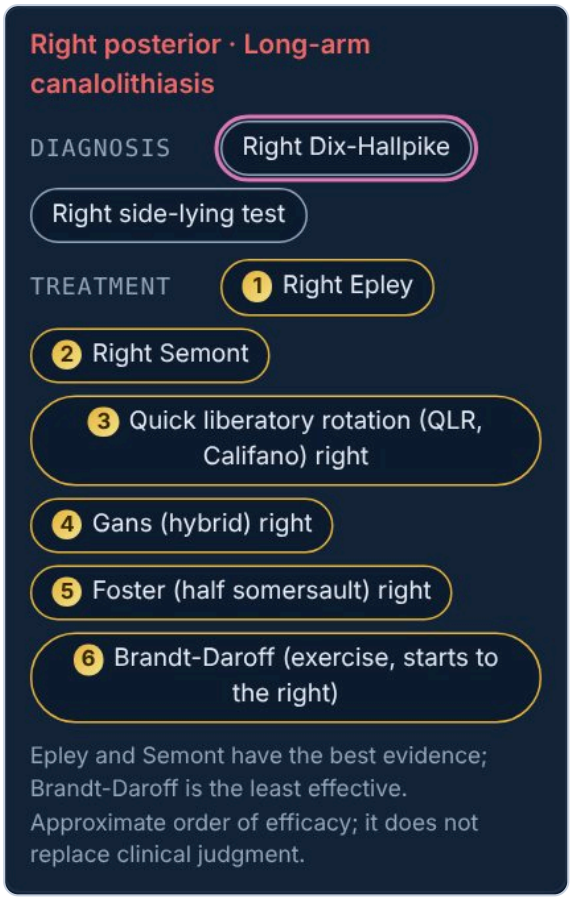


Figure 6. Diagnostic and treatment suggestions.

Test or maneuver

The drop-down list groups all tests and maneuvers. Those suggested for the case are marked: ● suggested diagnostic test and ★ suggested treatment with its number. Below, the step-by-step description of the chosen one shows the position and time of each step.

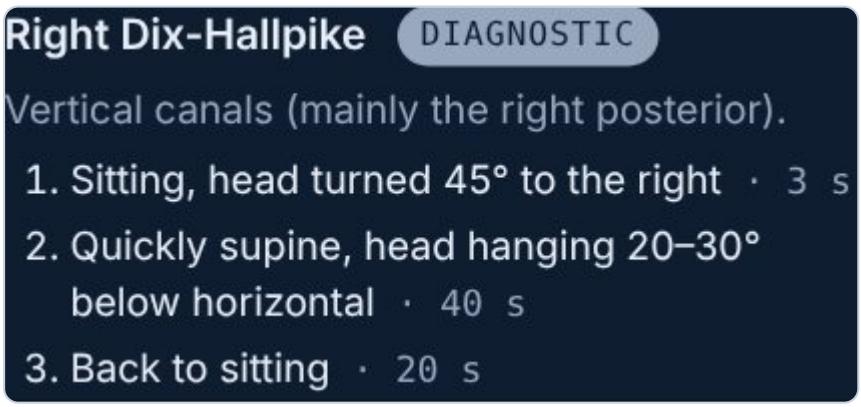


Figure 7. Step by step of the chosen maneuver.

Group	Tests and maneuvers
Diagnostic	Sitting (spontaneous nystagmus) · Right and left Dix-Hallpike · Midline head-hanging · Right and left side-lying test · Sitting → supine (lying-down) · Supine roll test to the right and to the left · Bow and lean (Choung).

Posterior canal	Epley · Semont · Gans (hybrid) · Foster (half somersault) · Demi-Semont (apogeotropic PC) · Brandt-Daroff; for each side.
Anterior canal	Modified Yacovino · Short Epley · Reverse Epley · Kim · Garaycochea (downbeat, each side).
Horizontal canal	Lempert 360° · Geotropic Gufoni · Vannucchi-Asprella · Forced prolonged position (geotropic and apogeotropic) · Modified Zuma · Gufoni-Appiani · Apogeotropic Gufoni nose-down · Zuma e Maia; for each side.
Mixed	Yacovino (deep head-hanging) · Quick liberatory rotation (QLR, Califano) · Hemi-precession (HC, uncertain side).

Running the maneuver



Figure 8. Run controls.

Control	Function
Run	Starts the chosen test or maneuver.
Pause / Continue	Stops and resumes the model clock.
Next step	Ends the current step and moves to the next one.
Mode	*Automatic*: each step lasts its time. *Step by step*: the patient stays in each position until you press Next step.
Speed	From 0.5× to 16×. 1× is real time.
Reset otoconia	Returns the otoconia to their initial position.
Reset when running	When on, each run starts from the untreated patient. When off, the progress of a treatment (detached deposit, partial release or arm change) is kept so you can continue it.

Free position

In **Free position** you place the patient in any posture: body (sitting, sitting on the edge, leaning, supine, side-lying, prone), head turn, flexion or extension and lateral tilt. The «This is how the patient will be positioned» preview shows the result before you apply it.

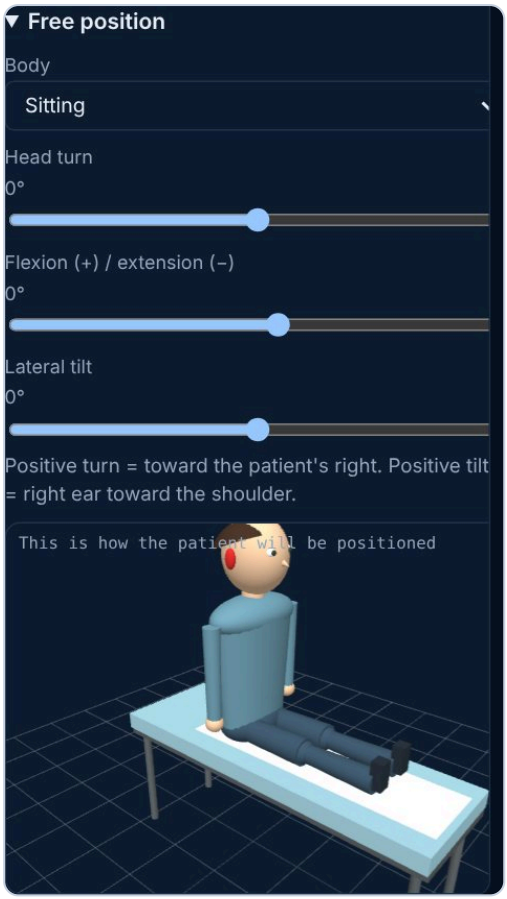


Figure 9. Free position of the patient.

3D view



Figure 10. Simulator during a right Dix-Hallpike with right posterior canalolithiasis.

The scene shows the patient's head with both labyrinths (right in red, left in blue), the otoconia, the cupula of each ampulla, the utricle, the saccule and the cochlea. The grey arrow shows gravity and the **Gravity** box describes it relative to the head. At the top left is the **video-oculography**: both of the patient's eyes with fast-phase arrows (purple torsional, yellow vertical and green horizontal). The **Patient** box shows the whole patient on the table.

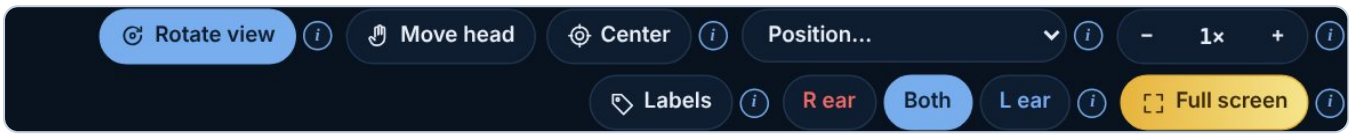


Figure 11. 3D view bar.

Control	Function
Rotate view	Dragging moves the camera (two fingers on a phone).
Move head	Dragging moves the patient within physiological ranges: the body sits up, lies down or turns, and the neck follows at its limit.
Center / 🎯	Returns the camera to the initial view.
Position…	Ready-made postures of tests and maneuvers (Dix-Hallpike, head-hanging, roll test, side-lying nose up or down, bow, lean, prone…).
− 1× +	Simulation speed.
Labels	Shows the names of the anatomical structures.
R ear · Both · L ear	Zooms the camera to one labyrinth or shows both.
Full screen	Wide view with the controls in a side or bottom panel.

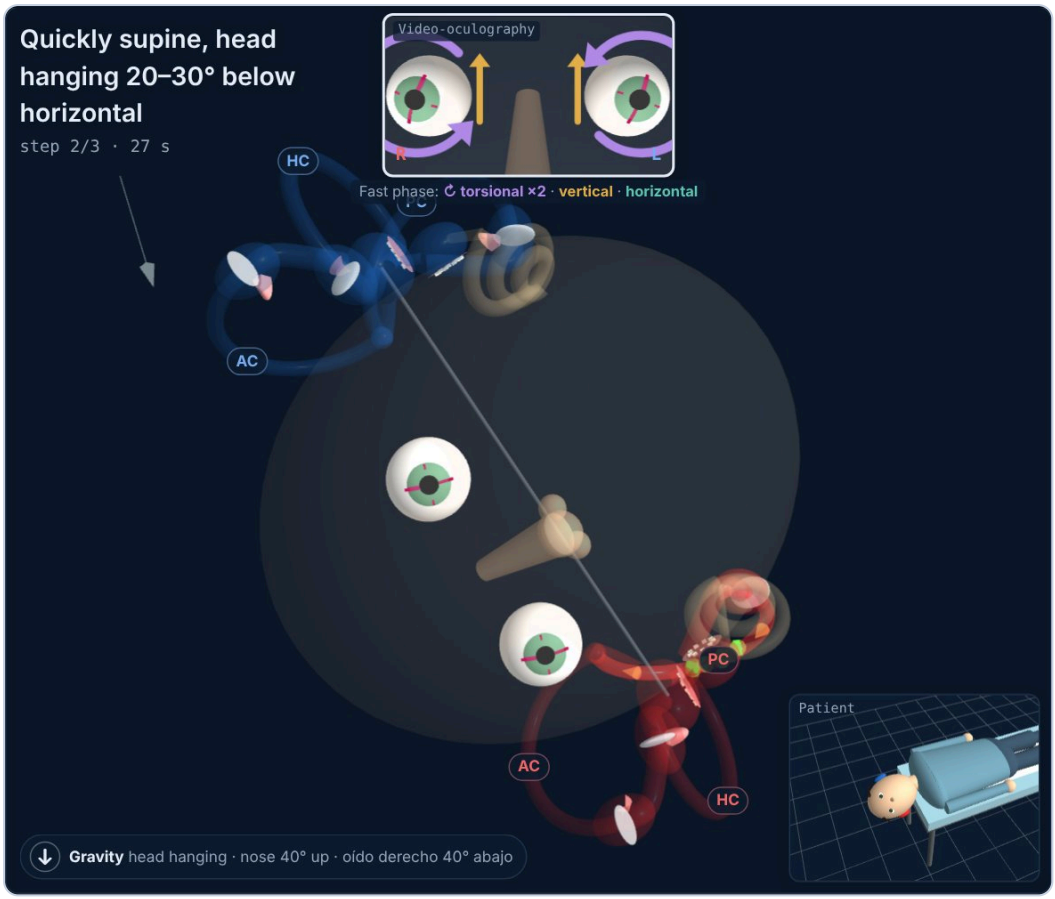


Figure 12. Anatomical labels.

Timeline

Below the view, the timeline shows each step as a segment that fills as it progresses. Tap a segment to go to that step; ⏮ goes back one step, ▶ runs or pauses and ⏭ moves forward.

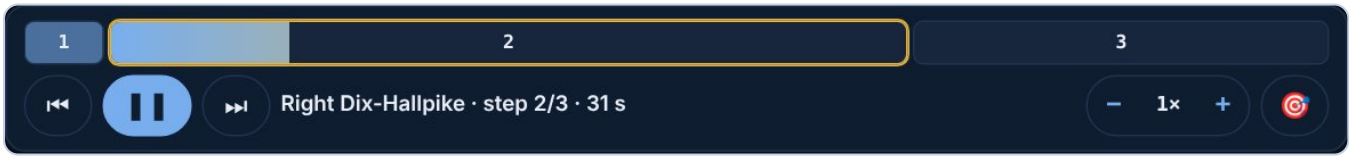


Figure 13. Maneuver timeline.

Status

Summarizes the canal, the variant, the direction of endolymph flow (ampullopetal or ampullofugal), whether it excites or inhibits, and how many otoconia have been released to the utricle.

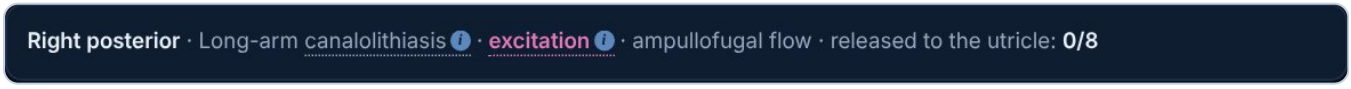


Figure 14. Status box.

Eye and expected nystagmus

The eye is shown as the examiner sees it: the patient's right side is on the left. The purple line on the iris marks the vertical meridian; its rotation is the torsion. The direction of the fast phase and the velocity of each component in °/s are shown. **Torsional emphasis ×2** only enlarges the drawing.

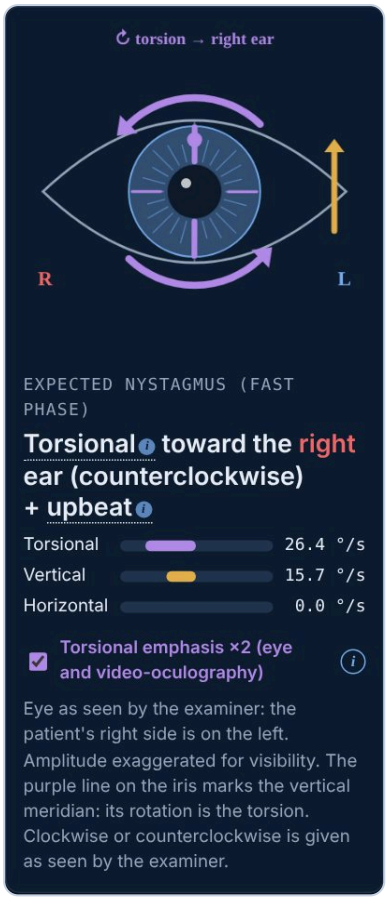


Figure 15. Eye and expected nystagmus.

Videonystagmography (VNG) recording

As in a VNG, the trace is the eye position over time: the ramp is the slow phase and the abrupt jump, the fast phase. Up means right (horizontal) and up (vertical).

- **VNG / SPV:** switches between eye position and slow-phase velocity.
- **torsional (3D VOG):** adds the torsional channel, as in 3D video-oculography.
- **10 s / 30 s:** width of the time window.
- ⏸ **Pause recording:** stops recording. While paused you can swipe over the recording or use the yellow bar to review up to 5 minutes back. ▶ **Resume recording** records live again.

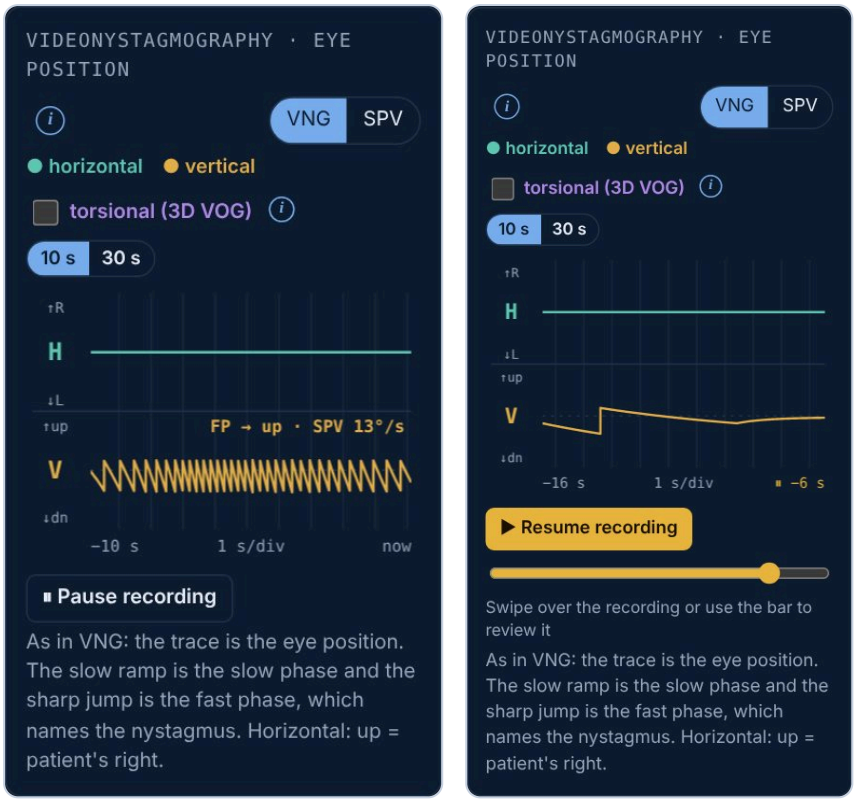


Figure 16. Live VNG recording and paused recording with the review bar.

Test report

At the end of each test, the report summarizes the nystagmus of each position: direction, latency, peak, duration and type (transient or persistent). **Session PDF** downloads all the tests of the session; **Clear** deletes the report.

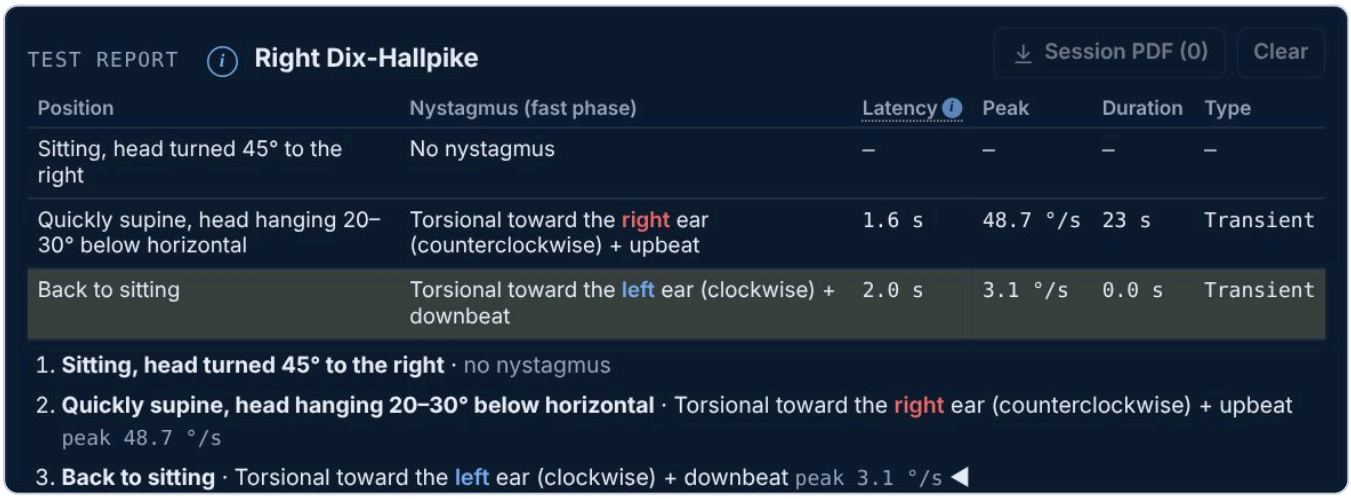


Figure 17. Test report.

Full screen

Enlarges the 3D view. In landscape the controls go to a side panel with shortcuts; in portrait, at the bottom. **Hide controls** leaves more room and **Exit** returns to the normal view.



Figure 18. Full screen.

Model notices

If the chosen test does not clinically explore the affected canal (for example, the left posterior canal in a right Dix-Hallpike), a notice appears: the model computes the physics in any position, but each canal must be explored with its own test.

5 Training module

The app presents a random clinical case from a bank of 38 verified cases (posterior, anterior and horizontal canal, with canalolithiasis and cupulolithiasis). The affected canal stays hidden until you check your answers.



Figure 19. Training mode.

How to use it

1. Read the **case presentation**, above the simulation.
2. Press ▶ **Run** to perform the case's test as many times as you like. Watch the eye, the video-oculography and the VNG recording.
3. With **Go to the questions** you answer: which test was done, which canals it explores, which nystagmus is seen, which canal is affected, the mechanism and the treatment, plus two vestibular physiology questions (basic and intermediate).
4. When you **check**, the canal and otoconia are revealed, with an explanation of each answer.
5. The score is stored on the device; reset it in **Settings**. **Exit training** returns to the free simulator.

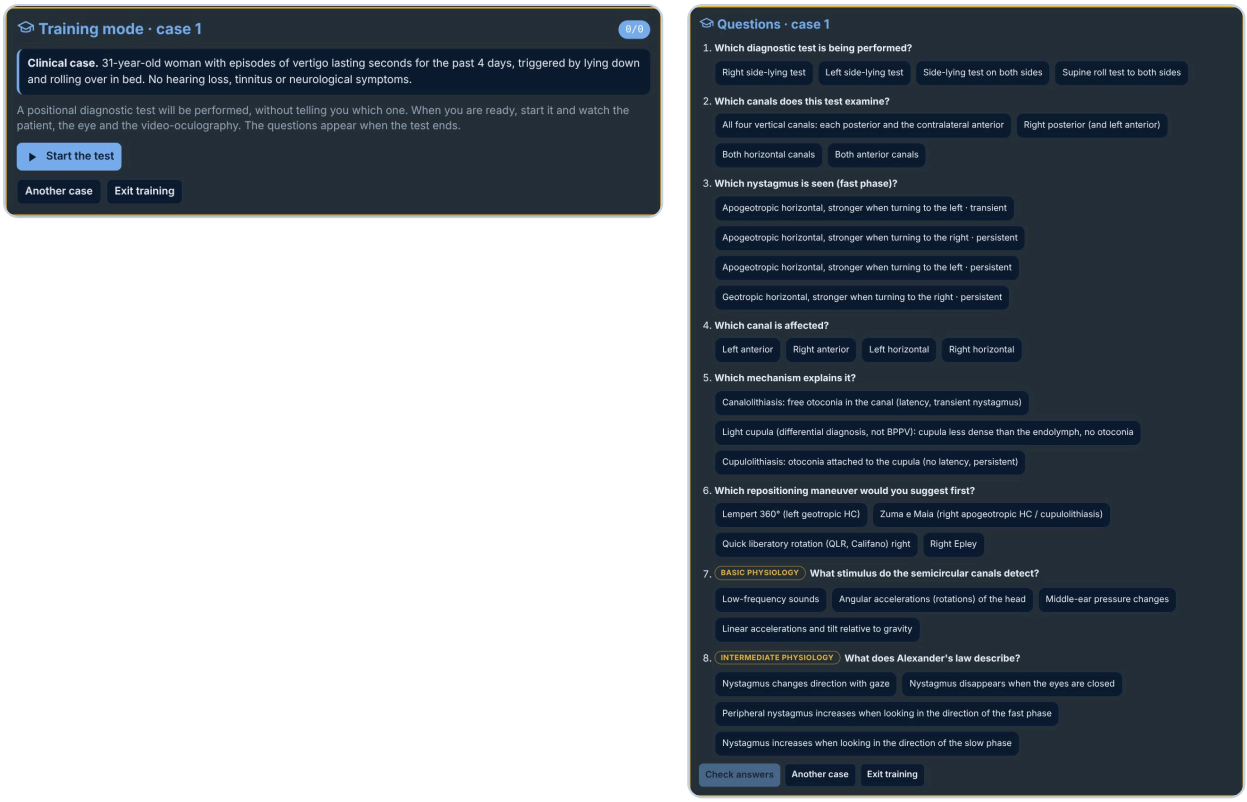


Figure 20. Case presentation and questions.

6 Clinical video module

Lets you analyze a real patient's video-Frenzel recording and compare it with the model. The video is processed on your device: it is not uploaded to the internet.

1. Video requirements

When you enter the module, the «Before loading your video» guide appears; you can reopen it with **Video requirements**. In summary:

- **A complete, correctly performed maneuver:** the model assumes the test was done as described in the literature. Tap each test to see the steps and timings the app uses and repeat them with your patient.
- **Timing of each position:** note when the patient reaches each position, hold each one at least 30–40 s (up to 60 s if you suspect cupulolithiasis) and record one continuous video per test.
- **Clear image:** whole pupil in focus, darkness without fixation, well-fitted goggles, few blinks.
- **Equipment and file:** one eye per video, know which one and whether the image is mirrored; .mp4 (H.264), ≥ 30 frames per second, ≥ 640×480, preferably under 5 minutes.
- **Patient:** vestibular suppressants, nystagmus fatigue on repetition and central warning signs.
- **Privacy:** videos without the face or identifying data and with consent.

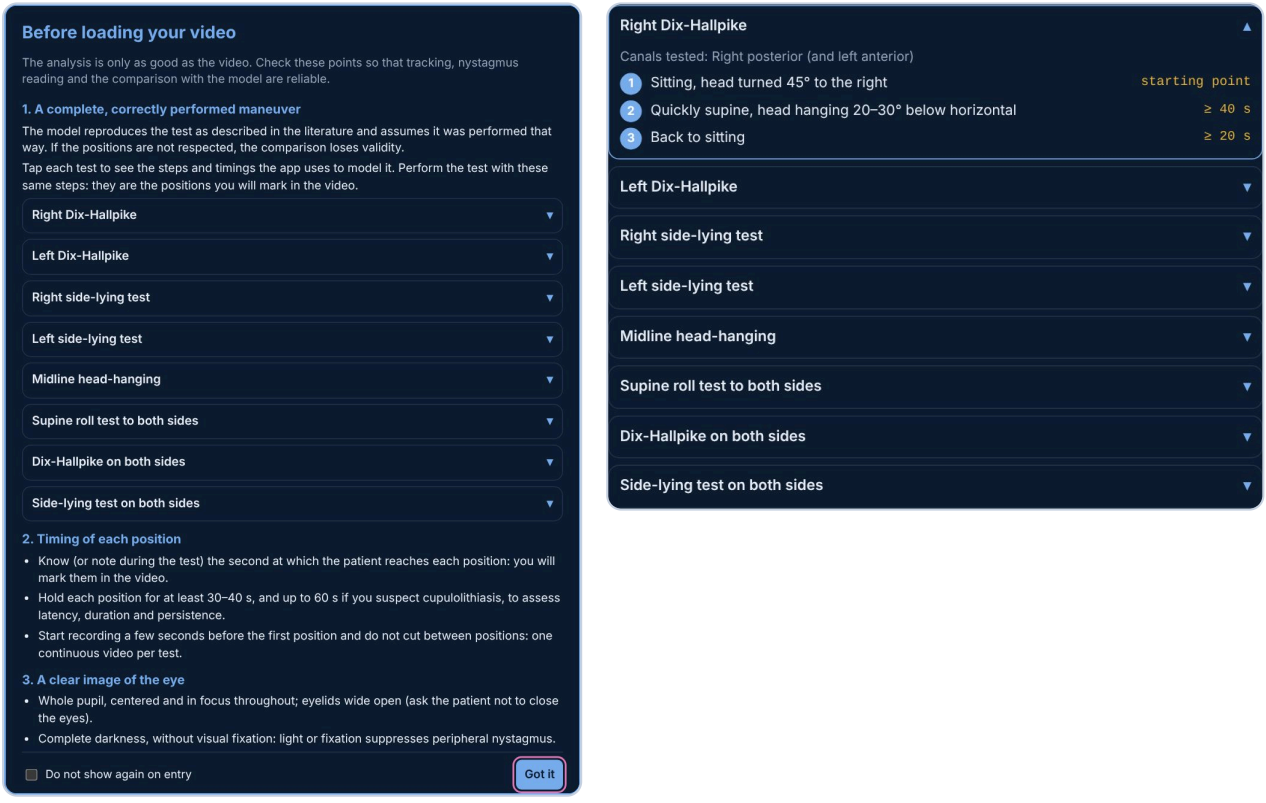


Figure 21. Requirements guide and steps of each test.

2. Load the video and the test data

Press **Choose video**. Then answer which test was performed, which eye was recorded and how the image looks (as the examiner sees it or mirrored).



Figure 22. Test data.

3. Mark each position

Play the video and, when the patient reaches each position, pause and press the yellow button. One tap per position is enough: each one lasts until the next mark. The ◀| and |▶ buttons step frame by frame and you can change the speed. The step list lets you go to a mark, correct it or delete it.

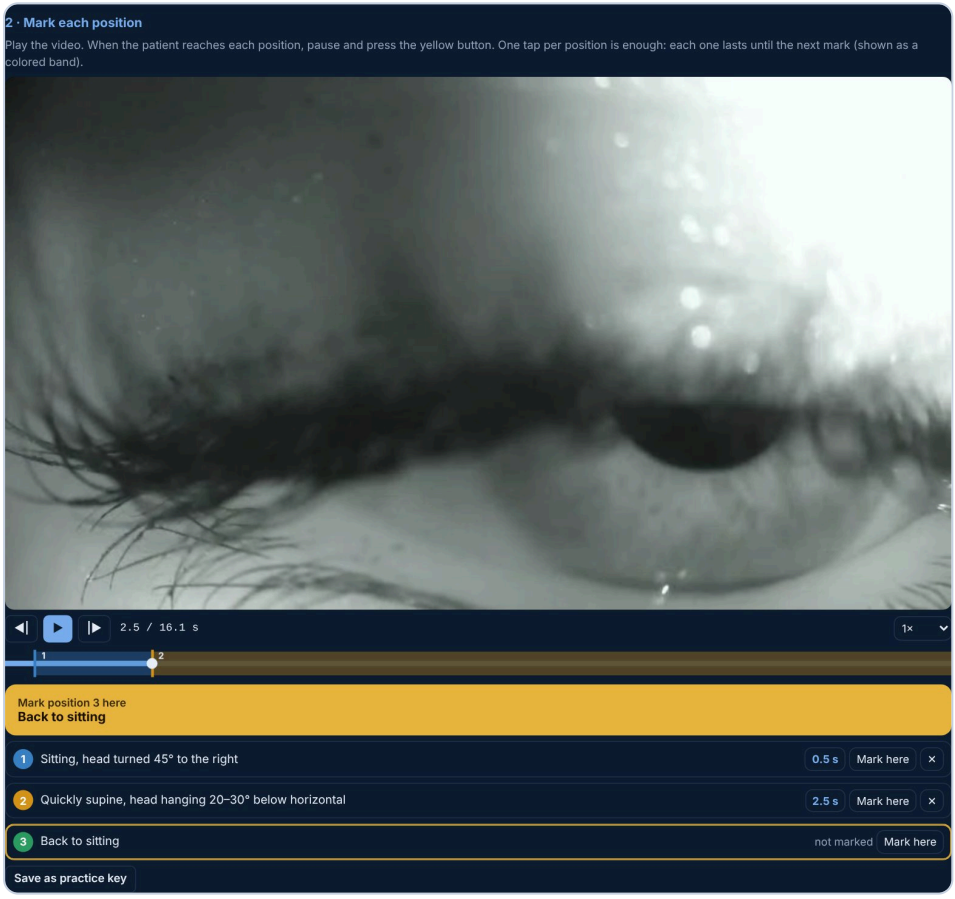


Figure 23. Marking positions.

Practice key: a teacher can save a key with the correct answers for a video. With **Load practice key** the student marks the video and the app grades the marks (± 2 s tolerance).

4. Analyze the pupil movement

Press **Analyze the video**. The first time you are asked to **tap the center of the pupil** in the video (yellow dot). Only the marked segments are analyzed; tracking is shown live as it progresses. If you leave the app, the analysis pauses and resumes when you return. **Cancel** stops it.



Figure 24. Analysis in progress.

When it finishes you see:

- **Quality:** percentage of frames with the pupil found and with torsion measured.
- **Traces:** horizontal, vertical and torsional, in detail (10 s) and for the whole video; tap the whole trace to jump to that moment.
- **Summary by position:** nystagmus direction, latency, duration and whether it is transient or persistent.
- **Arrows on the video:** while there is nystagmus, the video shows the fast phase with the video-oculography colors and the «Nystagmus recorded for the analysis» alert.

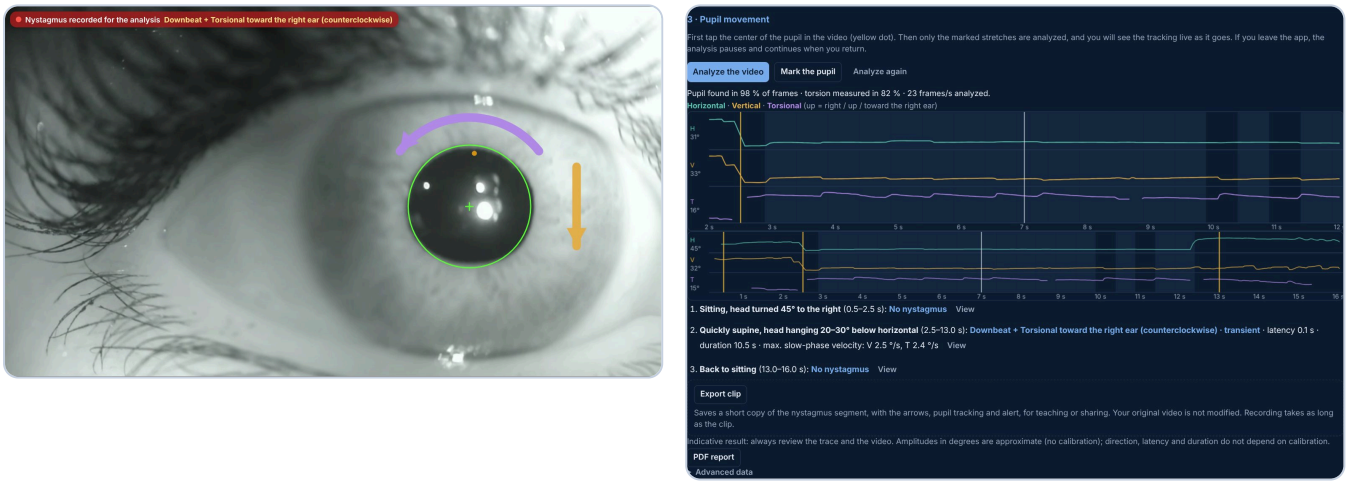


Figure 25. Arrows on the video and analysis results.

Button	Function
Analyze again	The first analysis of each video is saved; analyzing the same video with the same marks again gives an identical result. This button discards the saved analysis and tracks again.
Export clip	Saves a short copy of the nystagmus segment, with arrows, tracking and alert, for teaching or sharing. Recording takes as long as the clip; your original video is not modified.
PDF report	Test data, nystagmus by position, probable diagnosis, image of the 3D model next to the video at the moment of strongest nystagmus, and the VNG recording.

Advanced data (CSV)	Table with the eye position frame by frame, to check the numbers or for research.

5. Probable diagnosis

Calculate probable diagnosis simulates in the model each canal and variant that the test explores, with the same timings you marked, and ranks them by similarity to the measured nystagmus (direction of each component, persistence and latency). The three most likely options are shown with their percentage, the expected nystagmus and the suggested repositioning maneuver with alternatives. If the patient already had nystagmus in the starting position, the app flags it as possible spontaneous or pseudo-spontaneous nystagmus and gives it little weight.

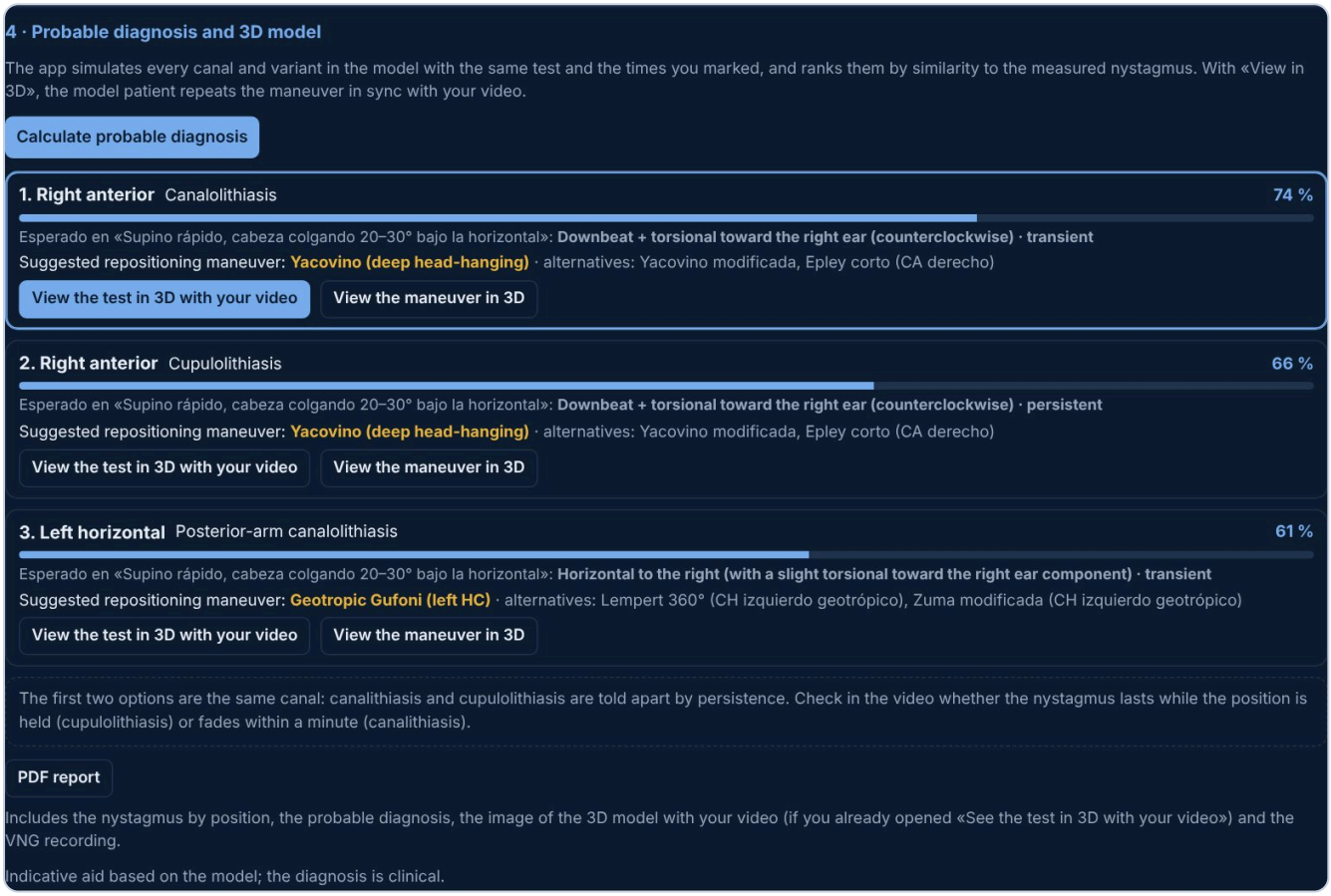


Figure 26. Probable diagnosis.

6. View the test in 3D with your video

The model patient repeats the test in sync with your video, which appears in a window (on a phone, below the 3D view; in full screen, compact in a corner). During the replay:

- **Eye: patient / model:** the 3D eye, the video-oculography and the VNG recording follow your patient's measured movement or the one computed by the model.
- The alert and the arrows appear over the eye in the video while there is nystagmus.
- Tapping a step, or ⏮, takes the video back to the start of that segment, even after it has ended.
- When the video ends, the VNG recording pauses so you can review it by swiping.
- **Locks 🔒:** during the replay, the controls that would change the patient's position or the otoconia are locked (Move head, Position..., Free position, canals and variants, quick scenarios,

suggestions, maneuver list, Run, Mode and Reset otoconia), so the model follows the recorded test. The view, speed, pause and steps remain available.

- **View the maneuver in 3D** shows the suggested repositioning maneuver; **Back to clinical video** returns to the module.



Figure 27. 3D replay in sync with the clinical video.

7. Saved session

The app saves the module's last session on your device: the video, the answers, the marks, the analysis, the diagnosis and the image of the 3D model. If you switch modules or close the app, when you come back «Previous session restored» appears and you do not need to analyze again. It is a single memory: it is replaced when you load another video, and **Start new** deletes it.

Limitations

Amplitudes in degrees are approximate (no calibration); direction, latency and duration do not depend on calibration. Torsion measured from the iris pattern is underestimated and misses fast phases. With few frames per second, blinks or a poorly visible pupil, tracking gets worse. The probable diagnosis is an indicative aid: the diagnosis is clinical.

7 Settings

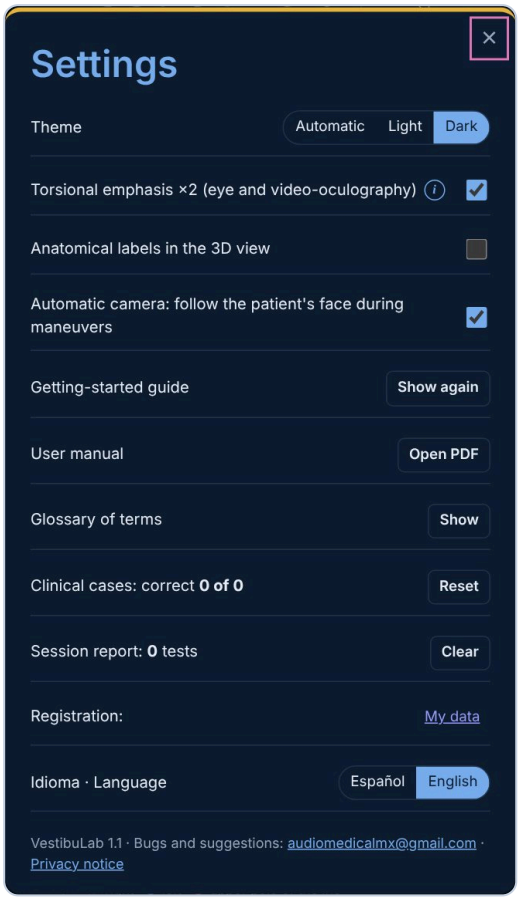


Figure 28. Settings.

Setting	Function
Theme	Automatic, light or dark.
Torsional emphasis ×2	Enlarges the drawn rotation of the eye and the video-oculography.
Anatomical labels in 3D	Shows or hides the names of the structures.
Automatic camera	The camera follows the patient's face during maneuvers.
Getting-started guide	Shows the guide again.
Glossary of terms	List of terms with their definition.
Clinical cases: score	Training score; **Reset** sets it to zero.
Session report	Number of tests in the report; **Clear** deletes it.
Registration · My data	Write to access, correct or cancel your data.
Idioma · Language	Spanish or English.
User manual	Opens this manual as a PDF.

8 How the model works and its limits

Scope: VestibuLab only simulates peripheral nystagmus caused by otoconia (BPPV due to canalolithiasis and cupulolithiasis) and, as a differential diagnosis, the light cupula. It does not recognize or simulate central nystagmus, spontaneous nystagmus from a vestibular lesion, congenital nystagmus or other types; that is why the «Scope of the model» notice appears when the app opens. If a real nystagmus does not behave as the model predicts, consider another cause.

- **Anatomy:** each canal is an arc in its anatomical plane measured by CT: the anterior at ≈38° from the sagittal plane, the posterior at ≈48° and the horizontal tilted ≈20° upward at the front. The LARP and RALP pairs are nearly parallel.
- **Otoconia:** calcium carbonate aggregates (≈2.71 g/cm³) in endolymph (≈1.0 g/cm³) that settle according to Stokes' law, with a friction threshold and a detachment time that reproduces clinical latency (1–5 s in the PC).
- **Ewald's laws:** the eye rotates in the plane of the stimulated canal; in the horizontal canal ampullopetal flow excites and in the vertical canals ampullofugal flow excites. Inhibition weighs 35 % of excitation.
- **Cupulolithiasis and light cupula:** the signal depends on gravity acting on the cupula, so the nystagmus is persistent. In the detachable variant the deposit comes loose with sustained traction, abrupt stops or head shaking.
- **Terminology:** Bárány Society criteria (von Brevern et al., 2015).
- **Head and neck:** active ranges of an adult (rotation 90°, flexion 50°, extension 60°, tilt 40°) and the limits of the table.
- **Limits:** canal switch, the real shape of the labyrinth, central adaptation and visual fixation are not modeled. It is a teaching tool that does not replace examination.

9 Glossary

3D video-oculography (VOG)
Recording of eye movement that adds the torsional component.

Ampulla
Dilation of each semicircular canal that holds the crista with the cupula.

Ampullofugal
Endolymph flow away from the ampulla. Excites the vertical canals and inhibits the horizontal one.

Ampullopetal
Endolymph flow toward the ampulla. It excites the horizontal canal and inhibits the vertical ones (2nd and 3rd Ewald laws).

Apogeotropic
Nystagmus whose fast phase beats away from the ground (toward the upper ear). In the horizontal canal it refers to the horizontal component; in the posterior canal, to the variant with torsion toward the upper ear and a downbeat component (otoconia in the non-ampullary arm).

BPPV
Benign paroxysmal positional vertigo: brief vertigo triggered by changes in head position, caused by otoconia in a semicircular canal.

Canalolithiasis
Loose otoconia moving inside the canal: nystagmus with latency, transient and fatigable. Term used in the Bárány Society classification; canalithiasis is also used and, in the Spanish SEORL-CCC guideline, conductolithiasis.

Common crus
Duct formed by the non-ampullary ends of the anterior and posterior canals before they reach the utricle.

Cupula
Gelatinous structure over the ampullary crista that bends with endolymph flow and stimulates the hair cells.

Cupulolithiasis
Otoconia attached to the cupula, which becomes denser than the endolymph and sensitive to gravity: nystagmus without latency that persists while the position is held. It is a BPPV type recognized by the Bárány Society (horizontal canal; posterior canal as an emerging form).

Cupulopathy
Umbrella term for heavy and light cupula: the density of the cupula no longer matches that of the endolymph and the canal responds to gravity.

Detachable
Model variant in which the deposit can detach from the cupula with sustained positions or liberatory maneuvers.

Dix-Hallpike

Diagnostic test for the posterior canal: head turned 45° toward the tested side and quick move to supine with the head hanging.

Downbeat

Fast phase downward.

Ewald's laws

Ewald's laws: the eye rotates in the plane of the stimulated canal; in the horizontal canal ampullopetal flow is excitatory, in the vertical canals ampullofugal flow; excitation is stronger than inhibition.

Excitation

Canal response when flow is in the excitatory direction; produces the strongest nystagmus.

Fast phase

Corrective component of nystagmus; by convention it names the direction.

Geotropic

Nystagmus whose fast phase beats toward the ground (toward the lower ear). In the horizontal canal it refers to the horizontal component on the roll test; in the posterior canal, to the torsional component: the typical form is geotropic because the upper pole of the eye beats toward the lower ear. It is not usually applied to the anterior canal.

Heavy cupula

Cupulopathy in which the cupula is denser than the endolymph: persistent apogeotropic nystagmus with a null point. Its best-recognized cause is cupulolithiasis, but it is also described without otoconia; they cannot be told apart by the nystagmus.

Inhibition

Canal response when flow is in the inhibitory direction; the nystagmus is weaker than with excitation.

LARP / RALP

Pairs of coplanar vertical canals: left anterior with right posterior, and right anterior with left posterior.

Latency

Time between reaching the position and the onset of nystagmus; typical of posterior canal canalolithiasis (1–5 s).

Light cupula

Cupulopathy without otoconia: the cupula is less dense than the endolymph. Persistent geotropic nystagmus with a null point toward the affected side; it does not respond to maneuvers.

Null point

Head turn (a few degrees from supine) at which the persistent nystagmus disappears because the cupula is aligned with gravity; it lies on the affected side in cupulolithiasis and in light cupula.

Otoconia

Calcium carbonate crystals of the utricular macula; when they detach and enter a canal they cause BPPV.

Persistent

Nystagmus that lasts as long as the position is held (more than 60 s); suggests cupulolithiasis or light cupula.

Pseudo-spontaneous nystagmus

Nystagmus that appears when sitting with the head upright, typical of the horizontal canal; it changes or disappears with head flexion.

Pupil tracking

Automatic location of the pupil center in each video frame to measure eye movement.

Roll test

Horizontal canal test: supine, head turned 90° to each side.

Side-lying test

Alternative to the Dix-Hallpike: sitting on the edge, head turned 45° to the opposite side, then lying on the tested side.

Slow phase

Eye movement produced by the vestibulo-ocular reflex, opposite to the fast phase; its velocity measures the intensity.

Slow-phase velocity (SPV)

Eye velocity during the slow phase, in °/s; it measures the intensity of the nystagmus.

Torsional

Rotation of the eye around its anteroposterior axis; named by which ear the upper pole of the iris moves toward. The clockwise or counterclockwise direction in parentheses is as seen by the examiner facing the patient: toward the right ear is counterclockwise and toward the left ear, clockwise.

Transient

Nystagmus that builds up, peaks and fades in less than 60 s; typical of canalolithiasis.

Upbeat

Fast phase upward.

Utricle

Otolith organ into which the canals open; its macula holds the otoconia.

Video-Frenzel

Goggles with an infrared camera that record the eye in darkness, without visual fixation.

Videonystagmography (VNG)

Infrared camera recording of eye position over time (horizontal and vertical).

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